MAHATMA GANDHI UNIVERSITY, KOTTAYAM MGU-UGP (HONOURS)
Regular EXAMINATION October 2025

Third Semester

Discipline Specific Elective (DSE) - MG3DSEFSQA01 - ESSENTIALS OF FOOD SAFETY (2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Part A
Short Answers
(2 Marks Each)
Answer in one or two sentences. Attempt any 10 questions.

1. Discuss on the economic benefits of ensuring food safety
   Answer: Ensures consumer confidence, reduces healthcare cost, promotes trade and productivity.

2. Write on bacterial food poisoning
   Answer: Illness due to ingestion of food containing harmful bacteria or toxins such as Salmonella or Staphylococcus.

3. Define high-risk foods with examples
   Answer: Foods that support rapid bacterial growth such as meat, poultry, dairy, and seafood.

4. State the methods to avoid the consumption of potentially contaminated food
   Answer: Proper cooking, hygienic handling, using safe water, avoiding cross-contamination.

5. Write on preventive methods for heavy metal poisoning.
   Answer: Source control, using clean raw materials, testing, and avoiding contaminated water.

6. Mention the features of Fasciola hepatica
   Answer: Liver fluke parasite, flatworm infecting cattle humans, transmitted via contaminated vegetation or water.
7. State why wooden equipment should be avoided in food premises
   Answer: Wood absorbs moisture, harbors bacteria, and is difficult to sanitize.

8. Define a food allergen.
   Answer: A substance in food that triggers an immune response, e.g., peanuts, milk, eggs.

9. Mention why supervision is necessary in allergen control
   Answer: To prevent cross-contact, ensure proper labeling, and monitor hygiene compliance.

10. State the reasons raw foods should be stored separately from cooked foods.
    Answer: To prevent cross-contamination, as raw foods may contain pathogens.

11. Define dry food storage
    Answer: Storage of foods at ambient temperature in dry, ventilated areas to prevent spoilage.

12. State the risk of thawing food at room temperature
    Answer: Bacteria multiply rapidly as outer layers warm before inner parts thaw.

Part B – Short Essay Questions (5 Marks Each)
Answer any 6 questions. Each question carries 5 marks.

13. Explain physical hazards with examples
    Answer: Physical hazards are foreign materials like glass, metal, or stones that can cause injury. Prevented by sieving, filtration, and detection methods.

14. Define HACCP, Critical Control Point, and Critical Limit
    Answer: HACCP is a preventive system ensuring food safety. CCP is a stage where control can be applied to prevent hazards. Critical limit is the maximum or minimum value to control the hazard.

15. Explain the control measures for preventing norovirus outbreaks.
    Answer: Proper handwashing, disinfection, excluding infected workers, and ensuring safe water use.

16. Explain the features of any three infectious bacterial species.
    Answer: Salmonella (poultry), Listeria (refrigerated foods), and E. coli (fecal contamination). Each causes gastrointestinal symptoms.

17. Describe measures to prevent cross-contamination during food handling.
    Answer: Separate raw/cooked foods, sanitize surfaces, maintain hygiene, and use different utensils.

18. Describe the source and role of dust and soil in physical hazards.
    Answer: Dust and soil carry microbes and heavy metals. Enter food during harvesting or storage; controlled by cleaning and air filtration.

19. Discuss safe practices for frozen food storage.

Answer: Maintain temperature below -18°C, proper packaging, avoid refreezing, FIFO system, and labeling.

20. Discuss the safety of vending machines used for food.
    Answer: Ensure temperature control, hygiene, clean water supply, regular servicing, and shelf-life monitoring.

Part C – Essay Questions (10 Marks Each)
Answer any 2 questions. Each question carries 10 marks.

21. Discuss the importance of protecting vulnerable populations from foodborne illnesses and describe the risks associated with consuming raw or undercooked foods.
    Answer: Vulnerable groups include infants, elderly, pregnant women, and immunocompromised individuals. Risks include bacterial infections and toxins. Protection involves proper cooking, hygiene, education, and regulatory measures.

22. Explain the sources, health hazards, and control measures of mycotoxins in food.
    Answer: Mycotoxins are toxic compounds from molds like Aspergillus. Found in cereals, nuts, and spices. Hazards include liver damage and cancer. Control through drying, storage, testing, and HACCP.

23. Explain the importance of training and supervision in chemical hazard prevention.
    Answer: Training ensures safe handling of cleaning agents and pesticides. Supervision monitors compliance and records. Prevents misuse, contamination, and accidents.

24. Discuss in detail the importance of packaging and wrapping methods in food storage, distribution, and safety.
    Answer: Packaging protects food from contamination and damage, maintains quality, extends shelf life, and aids distribution. Includes glass, plastic, and eco-friendly materials.